

Institute of Actuaries of India

Subject SP8 – General Insurance: Pricing

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) A stop loss responds to the aggregate of losses in the year (rather than individual large losses or large catastrophic losses). A view of the loss ratio can be derived from historical experience, suitably adjusted. This loss ratio would need to be modelled stochastically, to assess the volatility and hence the likelihood (and extent) of large aggregate losses. [1.5]

Alternatively, the losses may be modelled separately – for example, by splitting out attritional, large and catastrophe losses (or attritional and large/catastrophe).

1. The catastrophe loss could come from a vendor model (where applicable).
2. The large losses could come from a frequency and severity approach.
3. The attritional losses could be assessed using past historical attritional experience, with suitable adjusted.

This may produce a better model of the overall volatility. [1.5]

[3]

- ii) Below are Important considerations when pricing a stop loss include:
1. meeting risk transfer criteria (regulatory minima), ie any regulatory minimum transfer of risk
 2. the particular terms of the stop loss in question
 3. any inuring reinsurance
 4. the amount of the recoverable layer returned by the cedant.

[4]

[7 Marks]

Solution 2:

- i) For professional indemnity business, the limit purchased does not provide much information about the potential loss severity in the way that the insured values do for property treaties. So there is no gain in the analysis by switching from the original loss distribution X to the relative loss size distribution Y, as there is no reason to believe that Y will be the same for different risks. [1.5]

In professional indemnity business, no easily definable upper limit on the possible severity of loss to the original insured, which will often depend on court awards. Many factors influencing the limit actually purchased may not be connected with the potential loss severity (such as the insured's insurance budget, costs of cover, insured's risk aversion, market practice, cover purchased historically). [1.5]

Due to above reasons, ILFs are used in professional indemnity business.

[3]

ii)

Sum Insured (in Cr. INR)	Ground-up Loss amount (in Cr. INR)	Loss as % of Sum Insured
5.40	4.00	74.1%
10.00	2.40	24.0%
10.00	8.00	80.0%
9.50	7.50	78.9%
6.00	4.00	66.7%
5.00	2.50	50.0%
2.50	1.50	60.0%
10.00	4.00	40.0%
3.00	0.70	23.3%
10.00	7.60	76.0%

[2]

A % of Sum Insured (x)	B Total Value of losses ≤ x% of SI	C Losses > x% of SI (capped at x% of SI)	D Accumulated Losses at x% (B + C)	E Empirical Exposure Curve (D / Total Claims)
25%	3.10	14.60	17.70	41.9%
50%	9.60	21.70	31.30	74.2%
75%	19.10	22.13	41.23	97.7%
100%	42.20	-	42.20	100.0%

[4]

[Max 7]

[10 Marks]

Solution 3:

The mean and variance of the Gamma distribution

$$\mu(\Theta_t) = E(X_t | \Theta_t) = V_t / (V_t | \Theta_t) = \Theta_t \quad [1]$$

$$\sigma^2(\Theta_t) = V_t (\text{Var}[V_t | \Theta_t] = V_t * [V_t / (V_t | \Theta_t)^2]) = \Theta_t^2 \quad [1]$$

Similarly, using the formulae for the mean, variance and second moment of the lognormal distribution:

$$\beta = E(\mu(\Theta_t)) = E(\Theta_t) = 244.69 \quad [1]$$

$$\lambda = \text{Var}(\mu(\Theta_t)) = \text{Var}(\Theta_t) = 102880.65 \quad [1]$$

$$\phi = E(\sigma^2(\Theta_t)) = E(\Theta_t^2) = \text{Var}(\Theta_t) + (E(\Theta_t))^2 = (244.69)^2 + 102880.65 = 162754.79 \quad [1]$$

Using the Bühlmann-Straub formula for the credibility factor:

$$\text{The total number of policies} = V_t = 65+70+75+80+90+95 = 405 \quad [1]$$

$$Z_t = V_t / (V_t + (\phi / \lambda)) = 405 / (405 + (162754.79 / 102880.65)) = 99.61\% \quad [1]$$

$$X_6 = (S_1 + S_2 + S_3 + S_4 + S_5) / (V_1 + V_2 + V_3 + V_4 + V_5) = (33150 + 76500 + 22400 + 10800 + 48450) / 405 = 472.35 \quad [1]$$

The Bühlmann-Straub credibility premium per policy is

$$C^{BE} = Z_t X_t + (1 - Z_t) \beta = 99.61\% * 472.35 + (1 - 99.61\%) * 244.69 = 471.46 \quad [1]$$

$$\text{Risk premium for the total losses in year 2023 is } V_6 * C^{BE} = 471.46 * 100 = 47,146. \quad [1]$$

[10 Marks]

Solution 4:

The current rating factors seems sensible as it is expected risk to vary by these levels.

For example:

- Customers may have more liking of a particular type of vehicle compared to other areas e.g. more hatchbacks are in east, imported vehicle are more in west etc ..
- motor accidents may be more frequent where high commercial, industrial and economic activities are.

- motor accident costs higher where labour cost is higher also linked with type and make of model
- Few areas are more prone to theft due to more notorious activities
- malicious damage to cars probably higher in some areas

[4]

The proposal to use population density seems a level up in terms of sophistication in model and it is an objective extension to the current rating factors. [1/2]

A decision however needs to be made on the definition of population density as this requires some kind of partition of the country in the first place. [1/2]

Four levels of rating area, assuming each will have a large amount of historical exposure, will give credible estimates for the fitted model parameters. [1/2]

However, the current four levels will probably each contain a large range of varied experience e.g. there could be pockets in each current four levels who are at very low risk, and others will be very high. [1/2]

The four levels may not be able to partition the spectrum of risks appropriately e.g. some of the big cities in west behaves very similar to large cities in south and vice versa. [1/2]

Increasing the number of rating areas will therefore help resolve the issues above and lead to more accurate pricing provided that with 20 zonal groups instead of 4 credibility is not lost in our estimates this is where care needs to be taken in deciding the 20-population density bands. [1/2]

The choice of exactly 20 seems arbitrary and will also lead to a change in competitive position and reduce the risk of anti-selection and should help in increasing the profitability. [1/2]

Other considerations should also be giving to the IT systems, sales forces capabilities and customer adaptiveness. The accuracy of population density information should also be gauged either it's based on static vs dynamic, government agencies or private firm-based information etc. [1]

The Pricing Manager might want to consider whether separate rating area allocations should be derived for claim frequency and claims severity. [1]

The existence of a correlated factor in the model may limit the benefit of the proposed suggestion and introducing this may generate large swings in customers' premiums so may want to smooth the transition over time. [1/2]

[10 Marks]

Solution 5:

i) Gradient boosted model (GBM):

GBMs are decision tree-based models but share many mathematical and conceptual similarities with GLMs and are a natural extension of GLMs. GBMs are part of a class of algorithms called 'ensembles'. An ensemble algorithm combines multiple models together to generate a single algorithm that is more performant than the individual algorithms. In GBMs, each individual algorithm is a decision tree. A decision tree is an algorithm that uses a series of 'questions' to partition data and arrive at a solution.

[2]

Artificial neural network:

In an artificial neural network, each neuron takes inputs and applies a linear transformation, of the form $WX + B$, where each input, X , is multiplied by a weight, W , the resulting products are summed, and a bias term B is added. The weights and bias terms are all parameters that are learnt during the model fitting process. The output of this linear transformation then passes through a non-linear transformation, called an activation function, and the outputs of this activation function are the outputs of the nodes.

[2]

[4]

- ii) **Interpretability: extent to which cause and effect can be observed within the model.** [0.5]
Pricing actuaries, underwriters gain more comfort if they can understand why the model is showing the output it is showing. [1]

Portfolio managers can manage their portfolio better if they interpret the model better. To understand which factors are leading to a higher or lower premium rate, and how exactly it is doing it. [1]

Explainability: extent to which internal mechanics of a model can be explained in human terms. [0.5]

1. The key stakeholders are policyholders, intermediaries, management and regulator. These key stakeholders might want to gain some perspective as per their requirements
2. For example, a customer or an intermediary might want to know why the premium quoted is high or low compared to other insurers
3. Regulator might want to know the internal mechanics of the model to verify if there is any unfair discrimination or biases embedded within the model.

(1 mark each)

[6]

- iii) **Machine Learning techniques such as GBM, Random Forests are quite powerful and could lead to more accurate models than GLM.** [1.5]
Even if the insurer is relying on a GLM for pricing, these techniques can still help. For example:

1. Techniques such as Random Forest show the significant explanatory variables as a part of the output. This could help in improving the GLM model [1.5]
2. Identify segments where the GLM output is far off from the machine learning model. Understanding the reasons for deviation might help insurer in setting a more effective strategy for those segments [1.5]
3. Get more accurate estimates of loss ratio projections by running this model on business procured. For example, projection of loss cashflows for IFRS 17. [1.5]

[6]

[16 Marks]

Solution 6:

- i) OEP – the probability that the largest individual event loss in a year exceeds a particular threshold.
AEP – the probability that the aggregate losses from all loss events in a year exceeds a particular threshold.
Average Annual Loss: Expected claims; mean / average

[3]

- ii) $P = (L + 2) / 60\%$
where L represents annual expected loss for the treaty [1]

The value of L would be lower than AAL because the treaty has a limit of Rs. 116 Crore and not unlimited. The expected claims for this treaty would be lower than AAL, since events beyond 99.00% probability, the limit will be payable which is lower than the loss. [1]

- a) The limit of Rs. 116 Crore is applicable along with unlimited number of free instalments. In other words, the limit of Rs. 116 Crore is available for each event. [0.5]

- b) Therefore, the probability of the loss exceeding the limit for the largest individual event loss in a year is 99.00%. So we can approximately take 99.00% as probability of non-exceedance on AEP basis. The corresponding loss amount is Rs. 126.6 Crore [0.5]

Probability of Non-Exceed	AEP	Avg. Loss Amount in the layer (A)	Layer Probability (B)	Layer Loss Cost A*B
99.99%	736.5	736.5	0.01%	0.074
99.95%	540.1	638.3	0.04%	0.255
99.90%	430.9	485.5	0.05%	0.243
99.80%	318.3	374.6	0.10%	0.375
99.60%	215.2	266.75	0.20%	0.534
99.50%	188.3	201.75	0.10%	0.202
99.00%	126.6	157.45	0.50%	0.787
TOTAL				2.47

[1]

For the layer corresponding to probability of non-exceedance above 99%, the loss cost is 2.47 if there was no limit. With limit, we can take the loss cost to be 1% * Limit

$$= 1\% * 126.6 = 1.23$$

[1]

Impact of imposing 116 crore limit = 2.47 - 1.23 = 1.24 crore

Therefore, L = AAL minus 1.24 ~ 17.2 and premium = 32.0 crore

[5]

- iii) The revised premium would definitely be greater because higher limit means greater amount of expected claims. [1]

However, an increase in premium to the extent of the increase in expected claims is not sufficient. [0.5]

The reinsurer should also consider following in order to charge more premium:

- The variability is much more in the higher layers. Hence higher risk loading is needed for the layer 116 to 177 Crore. [1]
Typically, the variability is measured using measures such as Coefficient of Variation. The CoV for the layer 99% to 99.5% would be much more than CoV for layer 0% to 99%. [0.5]
- Due to higher variability and greater exposure, the Capital Requirement for the reinsurer might also go up significantly. [1]

[4]

iv)

- Insurance companies can monitor the business they write and the aggregate exposure to loss from extreme events. [1]
- Managing aggregate risk can minimize risk based capital requirement /optimal use of available capital [1]
- In addition, minimizing the concentration of risk may help reduce premium for catastrophe reinsurance costs [1]
- Some Catastrophe models can show how much the risk on its entire book increases when a specific risk is added to the existing book: in terms of AAL, OEP or AEP [1]
- Some catastrophe models can show how much premium changes based on changes to coverage such as limit amount. This way, these coverage and premium adjustments allow previously uninsurable properties to obtain coverage. [1]

6. Catastrophe models can be used to determine the impact of each rating factor, such as construction, year built, occupancy, and territory relativities [1]

[Max 5]

[17 Marks]

Solution 7:

- i) Rating factors:
- a. Manufacturer of vehicle / brand
 - b. Year of manufacture / age of vehicle
 - c. Model of vehicle
 - d. Type of vehicle
 - e. Odometer reading
 - f. Engine capacity / Battery capacity (EV)
 - g. Daily expected mileage
 - h. City/Address (residence + office)
 - i. Expected time of peak driving
 - j. Parking facility (street/garage)
 - k. Sum Insured (price of car)
 - l. Cover
 - m. Excess/deductible
 - n. Age of policyholder
 - o. Gender (if permissible)
 - p. Driver occupation
 - q. Past claims record of the driver
 - r. NCD level
 - s. Security features
 - t. Modifications, if any

[1/2 mark per point, **Max 4**]

- ii) Risk factors which are useful but not normally used
- a. Traffic density
 - b. Driving ability
 - c. Theft risk
 - d. Actual mileage

Rating Proxies used:

- Address (proxy for traffic density)
- Expected time of peak driving (proxy for traffic density)
- Age (proxy for driver ability)
- Occupation (proxy for driver ability)
- Past claims records (proxy for driver ability)
- Parking facility (proxy for theft risk)
- Security features (proxy for theft risk)
- Expected mileage (proxy for actual mileage)

Rating factors must be capable of being objectively measurable. In the cases above the risk factors cannot be rating factors as they are not objectively measurable and hence alternative rating factors are used.

[1/2 mark per point, **Max 6**]

- iii) Policy holder:
- a. +
 - i. only pay for cover as needed

- ii. potentially lower annual cost
- iii. good if drive infrequent long distances
- b. –
 - i. Inconvenience to opt for policy cover every day
 - ii. illegally driving without insurance cover if forget to arrange
 - iii. may not include theft / fire damage when not insured
 - iv. no other drivers allowed
 - v. above a certain number of days, the cost of cover may be more than a standard policy.
 - vi. garage restriction
 - vii. might lose NCD during the year (leading to higher cost)
 - viii. potential to be uninsured if annual mileage is >11k kms.

Insurer

- c. +
 - i. potential to penetrate niche market
 - ii. possibility of applying rate changes during the policy year
 - iii. might attract good risks / educated / professionals due to complication
- d. –
 - i. Marketing may be tough due to added complexity of the policy
 - ii. Might be administratively expensive
 - iii. high startup tech costs (app / development, etc)
 - iv. possible disputes over cover if crash occurs when not covered
 - v. higher risk of fraudulent claims
 - vi. increased moral hazard
 - vii. higher tendency for adverse selection
 - viii. potential dispute if annual actual mileage is more than 11,000kms

[1/2 mark per point, **Max 10**]

iv)

Challenges:

- a. Since the insurer is new to motor policy, they will not have any own data....
- b.therefore need to buy in data or run a pilot
- c. no data on the number of days drivers use their cars so daily rate is not known with certainty.
- d. people may not be able to ascertain their expected mileage while buying a new car.
- e. The expenses and administration costs will not be known with certainty.
- f. There may not be any similar existing product, so market benchmark might not be possible.
- g. There may be regulatory restrictions.
- h. There may be errors in data collection on daily basis (for instance, server/syncing issues) and this may distort the daily pricing.
- i. Different cars may be calibrated differently, and this may impact the telematics data collected. The actuary must allow for these while analysing the inputs
- j. For instance, a smaller engine car might be accelerated more aggressively compared to car with larger engine.
- k. It may be a challenge to determine the penalty / deductible for non-authorized garage servicing cost in absence of any historical data. The actual cost for servicing non-authorized garages might be higher than the expected additional cost (due to frauds, etc.)

[1 mark per point, **Max 10**]

[30 Marks]
